

NIRS Signal Analysis for Contralateral Activation during Motor Imagery

Future Applications in Brain-Computer Interfaces (BCIs)

Group 15

Abstract—This study investigates contralateral brain activation using Near-Infrared Spectroscopy (NIRS) during motor imagery (MI) tasks of hand movements. Twenty-nine healthy adults performed left- and right-hand MI tasks across three sessions. NIRS data were acquired, filtered, and analyzed for changes in oxygenated hemoglobin (O_2Hb). Topographical maps and statistical analysis were conducted to observe differences between contralateral and ipsilateral hemispheric activations. Results revealed weak O_2Hb changes with no statistically significant differences between hemispheres, despite a visual trend suggesting increased contralateral activation in later sessions. These findings highlight challenges in detecting robust MI-related brain activity using NIRS and highlight the importance of optimizing experimental protocols and participant training to enhance future applications, particularly in Brain-Computer Interfaces (BCIs).

Keywords—Motor imagery, NIRS, contralateral activation, Brain-Computer Interface, hemodynamic response.

I. INTRODUCTION

Near-Infrared Spectroscopy (NIRS) is a non-invasive technique widely used to study brain activity. Brain activity leads to an increase in oxygen consumption, which is accompanied by an increase in cerebral blood flow due to neurovascular coupling. This leads to a change in the local oxygenated hemoglobin (O_2Hb) and deoxygenated hemoglobin (HHb) which can be detected by fNIRS [1]. In particular fNIRS can be used to explore brain activation during motor imagery tasks: research has shown that changes in $[O_2Hb]$ levels can reliably reflect the activation of motor cortices during imagined movements [2]. This study aims to evaluate if significant contralateral brain activation can be observed during motor imagery tasks using fNIRS. Specifically, the study focuses on identifying consistent patterns of contralateral activation across different sessions and subjects, in order to advance toward the development of robust NIRS-based Brain-Computer Interfaces (BCIs).

II. MATERIALS AND METHODS

A. Participants

Twenty-nine subjects (healthy adults) were recruited. They were divided into thirteen males and sixteen females. The mean year of birth of the participants is 1988, with a standard deviation of 4 years. Of the twenty-nine participants, twenty-eight were right-handed and one was left-handed. All subjects were naive to the MI experiment, and they were instructed to perform haptic motor imagery.

B. Data Acquisition

NIRS data was collected by NIRScout (NIRx GmbH, Berlin, Germany) at 10 Hz sampling rate [3]. Each adjacent source detector pair creates one physiological NIRS channel. Fourteen sources and sixteen detectors resulting in thirty-six physiological channels were placed at frontal (nine channels around Fp1, Fp2, and Fpz), motor (twelve channels around C3 and C4, respectively) and visual areas (three channels around Oz). The inter-optode distance was 30 mm. Continuous NIRS light intensity data were acquired at two different wavelengths: 760 nm and 850 nm.

C. Experimental Procedure

NIRS data was collected in an ordinary bright room. The subjects sat on a comfortable armchair in front of a 50-inch white screen. The distance between their heads and the screen was 1.6 m. They were asked not to move any part of the body during the data recording. The experiment consisted of three sessions of left and right-hand motor imagery (MI). Each session comprised a 1 min pre-experiment resting period, 20 repetitions of the given task and a 1 min post-experiment resting period.

D. Task

The task started with 2 s of a visual introduction of the task, followed by 10 s of a task period and resting period which was given randomly from 15 to 17 s. At the beginning and end of the task period, a short beep (250 ms) was played. For the visual instruction, a black arrow pointing to either the left or right side appeared at the center of the screen for 2 s. The arrow disappeared and then a black fixation cross was displayed during the task period. The subjects were asked to imagine hand gripping (opening and closing their hands) at a 1 Hz pace. Motor imagery was performed continuously over the task period. The task period was finished with a 'STOP' displayed for 1 s on the screen. The fixation cross was displayed again during the rest period and the subjects were asked to gaze at it to minimize their eye movements. This process was repeated twenty times in a single session (ten trials per condition in a single session; thirty trials in the whole sessions). In a single session, motor imagery tasks were performed based on ten subsequent blocks randomly consisting of one of two conditions: either first left and then right-hand motor imagery or vice versa.



Fig. 1. Schematic sequence diagram of the experimental task sequence.

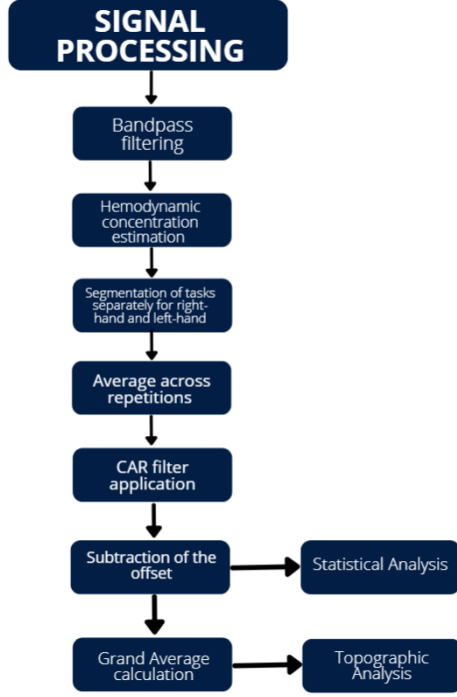


Fig. 2. Comprehensive workflow of the signal processing pipeline.

E. Signal processing

A fourth-order Butterworth bandpass filter was applied to the raw NIRS signals. The filter was set with a lower cutoff frequency of 0.01 Hz and an upper cutoff frequency of 0.09 Hz. Changes in oxygenated hemoglobin and deoxygenated hemoglobin concentrations were computed using the modified Beer-Lambert law (1). The absorption coefficients for the wavelengths of 760 nm and 850 nm were taken from published data [4].

$$\begin{cases} \Delta A_{\lambda_1} = \Delta c^{O_2Hb} \cdot \alpha_{\lambda_1}^{O_2Hb} \cdot d \cdot B + \Delta c^{HHb} \cdot \alpha_{\lambda_1}^{HHb} \cdot d \cdot B \\ \Delta A_{\lambda_2} = \Delta c^{O_2Hb} \cdot \alpha_{\lambda_2}^{O_2Hb} \cdot d \cdot B + \Delta c^{HHb} \cdot \alpha_{\lambda_2}^{HHb} \cdot d \cdot B \end{cases} \quad (1)$$

Task-related segments were extracted from the data based on onset markers provided. Each task included a 2-second preparation period followed by a 10-second motor imagery phase and a resting period of 15 seconds. These segments were separately labeled for right-hand and left-hand motor imagery tasks. The signals obtained during the task period were averaged across the 10 repetitions in a session, separately for each hand's motor imagery and for both $\Delta[O_2Hb]$ and $\Delta[HHb]$. A Common Average Reference (CAR) filter was applied to remove physiology-based systemic interference and, the task-related changes of oxy-Hb and deoxy-Hb were referred to the baseline value that was set as the value of the signal at the beginning of the task. The Grand-Average of $\Delta[O_2Hb]$ and $\Delta[HHb]$ signals was computed across all subjects for the three sessions for right-hand and left-hand motor imagery tasks. These averages were used to analyze the spatial distribution of brain activation in terms of peak amplitude, in particular the peaks amplitudes from the grand average signals were mapped topographically focusing on motor areas (e.g channels near C3 for the left hemisphere and channel near C4 for the right hemisphere).

F. Statistical Analysis

To enhance robustness, a statistical analysis was performed at the individual subject level, and it was performed to determine if significant contralateral activations could be observed in the motor cortex during motor imagery tasks. For each subject, the peak amplitudes of $\Delta[O_2Hb]$ signals were extracted from selected channel triplets to focus on the right and left hemispheres: C1C3, FC3C3, CP3C3 (channels 27, 29 and 33) for the left hemisphere; C2C4, FC4C4, CP4C4 (channels 22, 17, 20) for the right hemisphere. The peaks amplitudes from the triplets were averaged to obtain a single mean value per hemisphere for each subject and for right- and left-hand imagery. A Kolmogorov-Smirnov test was performed beforehand to verify the normality of the data distributions. The data did not follow a normal distribution and for this reason the Wilcoxon signed-rank test, a non-parametric alternative to the paired t-test, was chosen to analyze the median values of the activations of the 2 hemispheres among all the participants. So, for both left and right-hand motor imagery, a paired Wilcoxon signed-rank test was performed to compare the contralateral and ipsilateral hemispheric values. Statistical significance was determined using a p-value threshold of 0.05.

III. RESULTS

The distribution of $\Delta[O_2Hb]$ peaks amplitudes was analyzed using the Grand-Average signals for left motor imagery and right motor imagery across the three different sessions. To enable a direct comparison, the same limits were set for the colormap in all topographical maps. Topographic distributions can be seen in Fig. 3. For the first session, no clear visual evidence of contralateral activation was observed in either hemisphere. However, in sessions two and three, a more pronounced contralateral activation pattern becomes evident for both left-hand motor imagery and right-hand motor imagery. In general, the activation values observed during motor imagery tasks are very small in comparison to the values that have been observed in real motor movements [5]. To complement the visual analysis, a statistical analysis was performed to verify if a significant difference exists between contralateral and ipsilateral hemispheric activations. While the topographical maps based on grand average signals suggested no clear evidence of contralateral activation, a quantitative statistical approach was employed on individual subject data. The p-values obtained for each condition are shown in Tab. 1. The results confirm that no statistically significant difference was observed between contralateral and ipsilateral hemispheric activations across all sessions for

	p-values	
	<i>RMI</i>	<i>LMI</i>
Session 1	0.84	0.97
Session 2	0.27	0.67
Session 3	0.46	0.30

Tab. 1. p-values obtained from the statistical analysis for each session of right motor imagery and left motor imagery.

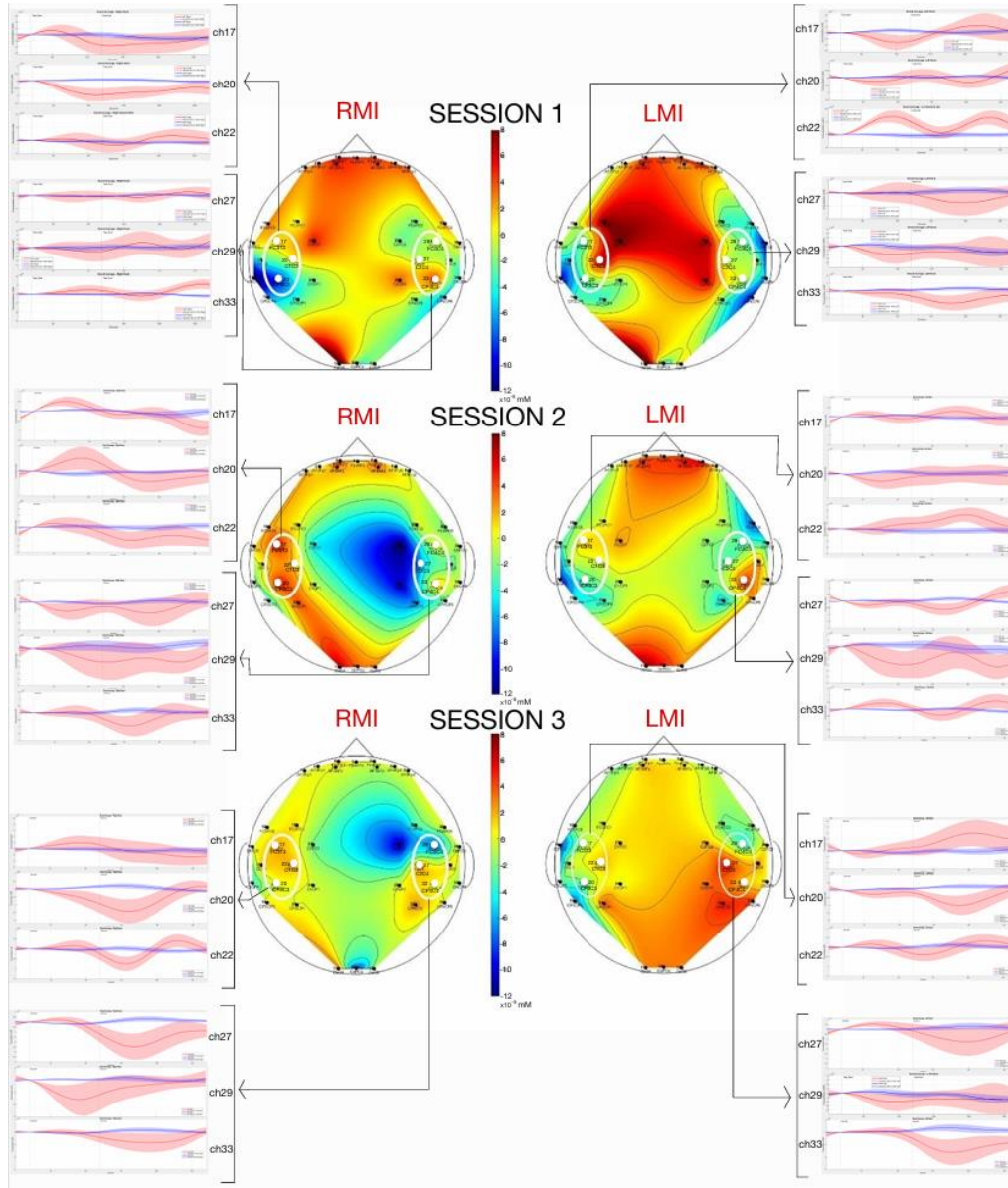


Fig. 3. Topographic maps of peaks amplitudes of relative concentration of oxygenated hemoglobin ($\Delta[\text{O}_2\text{Hb}]$) Grand-Average signals during right- and left-hand motor imagery (MI) tasks for the three sessions. The circled triplets of channels (e.g., C1C3, FC3C3, CP3C3 for the left hemisphere; C2C4, FC4C4, CP4C4 for the right hemisphere) indicate representative areas for contralateral and ipsilateral hemispheric activity. The plots that are shown are the Grand-Average signals of $\Delta[\text{O}_2\text{Hb}]$ and $\Delta[\text{HHb}]$ of the specific channels with the superimposition of the error bands.

both left- and right- hand motor imagery. The results are also shown in Fig. 4 where the mean amplitudes between subjects of $\Delta[\text{O}_2\text{Hb}]$ peaks for contralateral and ipsilateral hemispheres are compared across the three sessions for left-hand and right-hand motor imagery tasks. The bar plots illustrate the variability in the observed amplitudes, with error bars representing the standard deviations across subjects. In Tab. 2. there is a detailed numerical summary of these mean amplitudes and their standard deviations. For both conditions, no consistent trend favoring contralateral activation is evident. Specifically, in session one, for the right-hand motor imagery there is a contralateral deactivation and an ipsilateral activation and for left-hand motor imagery there is a small difference between contralateral and ipsilateral activation. In sessions two and three, although slight increases in contralateral amplitudes can be observed, the high variability suggests these differences are not statistically robust, as confirmed by the p-values in Tab. 1.

IV. DISCUSSION

From the topographical maps of $\Delta[\text{O}_2\text{Hb}]$ peaks amplitudes that are shown in Fig. 3. we can see that the observed values are quite small (order of $\sim 10^{-5}$ mM), and it is because it is just imagery. In the first session there is no clear visual evidence of contralateral activation, while in the second and third sessions a more pronounced contralateral activation pattern appeared to emerge for both motor imagery tasks. This visual trend suggests that participants may have improved their motor imagery performance over repeated sessions, potentially leading to more defined cortical responses. Participants, initially naive to motor imagery, could have improved their ability to perform the task through repeated practice, leading to a clearer cortical response. This result is consistent with literature suggesting that motor imagery

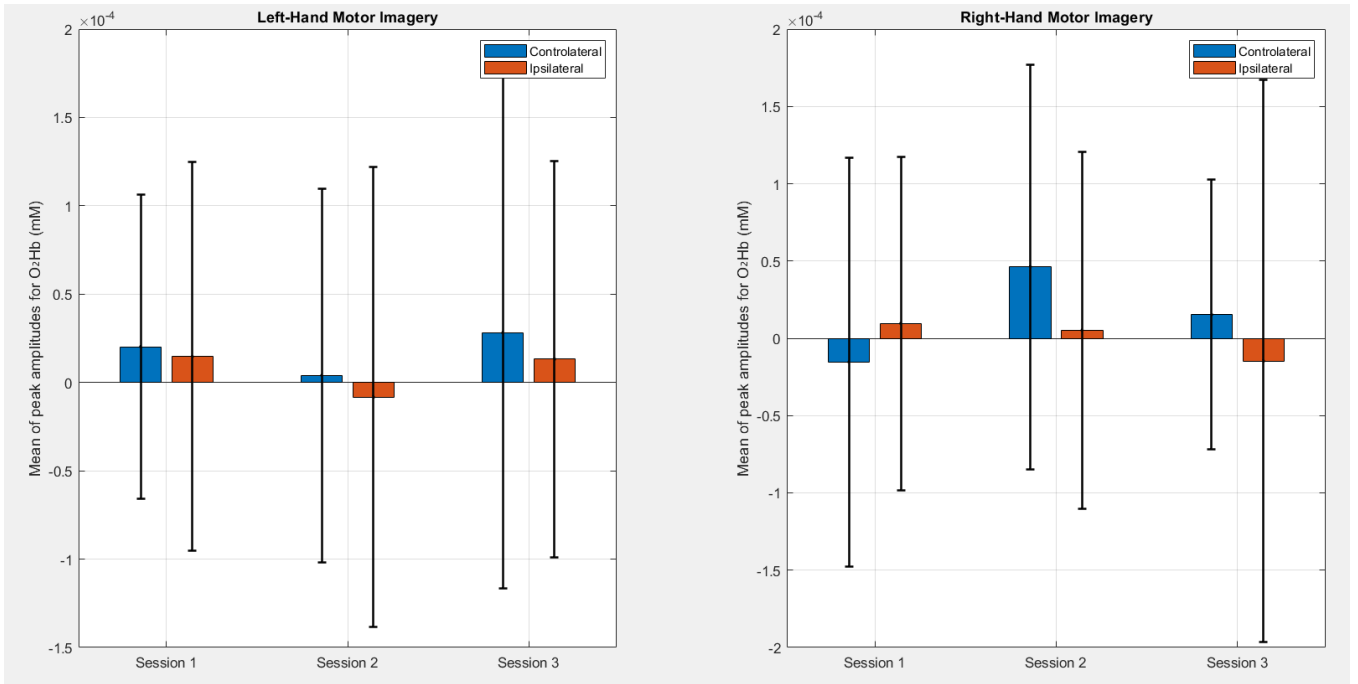


Fig. 4. Comparison between the mean of $\Delta[\text{O}_2\text{Hb}]$ peaks amplitudes between subjects for specific motor contralateral and ipsilateral channels during left-hand motor imagery (left panel) and right-hand motor imagery (right panel). The data represent contralateral (blue) and ipsilateral (orange) channels relative to the imagined motor side, shown for three distinct experimental sessions (Session 1, 2, and 3). Error bars indicate standard deviation, highlighting variability across subjects.

	$\Delta[\text{O}_2\text{Hb}]$ (mM)			
	RMI		LMI	
	Contralateral	Ipsilateral	Contralateral	Ipsilateral
Session 1	$(-1.53 \pm 13.23) \cdot 10^{-5}$	$(9.65 \pm 107.85) \cdot 10^{-6}$	$(2.04 \pm 8.60) \cdot 10^{-5}$	$(1.46 \pm 11.00) \cdot 10^{-5}$
Session 2	$(4.62 \pm 13.09) \cdot 10^{-5}$	$(5.17 \pm 115.47) \cdot 10^{-6}$	$(4.01 \pm 105.87) \cdot 10^{-6}$	$(-8.29 \pm 130.31) \cdot 10^{-6}$
Session 3	$(1.54 \pm 8.71) \cdot 10^{-5}$	$(-1.47 \pm 18.18) \cdot 10^{-5}$	$(2.81 \pm 14.46) \cdot 10^{-5}$	$(1.32 \pm 11.20) \cdot 10^{-5}$

Tab. 2. The table shows the mean between subjects and standard deviation (SD) of $\Delta[\text{O}_2\text{Hb}]$ peak amplitudes (in mM) for contralateral and ipsilateral motor channels during motor imagery of the left and right hand. The values are presented for each of the three experimental sessions. The "Contralateral" column represents data recorded from the side opposite to the imagined movement, while the "Ipsilateral" column represents data recorded from the same side as the imagined movement.

performance and associated brain activation improve with training [6]. The statistical results support the observations made in the topographical analysis. The Wilcoxon signed-rank test revealed no significant difference between contralateral and ipsilateral hemispheric $\Delta[\text{O}_2\text{Hb}]$ values for both left- and right- hand motor imagery tasks. While the visual analysis shows a trend toward contralateral dominance, the lack of statistical significance suggests that the differences between contralateral and ipsilateral activations are minimal and may not be robust enough to distinguish hemispheric dominance clearly. These findings align with previous studies reporting challenges in achieving statistically significant motor cortex activation during imagined movements, particularly when compared to real motor execution tasks [7]. The small magnitude of activation observed during motor imagery tasks can be attributed to the inherently subtle hemodynamic changes associated with imagined movements. Unlike real motor execution, which involves strong cortical activation and neurovascular responses, motor imagery tasks elicit weaker neural signals, resulting in smaller $\Delta[\text{O}_2\text{Hb}]$ changes detectable by NIRS. Moreover, the large variability in the observed data may also be influenced by individual differences such as hair color and hair length, as fNIRS

sensitivity can be affected by these factors. For example, darker hair or longer hair may attenuate the light signal, reducing the accuracy of measurements [8]. This difference highlights the need for refined task protocols, better participant training, or advanced signal processing techniques to enhance the sensitivity of fNIRS in detecting motor imagery-related brain activity. To address these challenges, future work should focus on increasing the sensitivity of fNIRS signal detection, introducing a neurofeedback to enhance brain activation and conducting studies to explore learning effects and improvements in motor imagery ability over time. Addressing variability due to individual physical characteristics, such as optimizing probe placement or using adaptive calibration methods, can also be crucial for advancing toward future applications in Brain-Computer Interfaces.

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